

Inquiry Service Documentation

Overview

The Inquiry Service handles customer inquiries about books, availability, pricing, and custom orders. It manages the complete inquiry lifecycle from creation to resolution, including owner responses and status tracking.

Service Responsibilities

- Inquiry creation and management
- Inquiry status tracking
- Owner response handling
- Inquiry filtering and search
- Inquiry statistics and reporting
- Email notifications for inquiries

API Endpoints

1. Create Inquiry

Endpoint: POST /api/inquiries

Description: Create a new inquiry

Headers: Authorization: Bearer {token}

Request Body:

```
{
  "type": "BOOK_AVAILABILITY",
  "subject": "Availability of Mathematics Class 10",
  "message": "Do you have Mathematics for Class 10 by Dr. Ramesh Kumar in stock?",
  "productId": 1
}
```

Response: 201 Created

```
{
  "id": 1,
  "type": "BOOK_AVAILABILITY",
  "subject": "Availability of Mathematics Class 10",
  "message": "Do you have Mathematics for Class 10 by Dr. Ramesh Kumar in stock?",
  "productId": 1,
  "productName": "Mathematics for Class 10",
  "status": "PENDING",
  "userId": 5,
  "userName": "John Doe",
  "createdAt": "2025-12-23T10:00:00Z",
  "responses": []
}
```

Authorization: Requires authentication (CUSTOMER)

Inquiry Types:

- BOOK_AVAILABILITY : Inquiry about book availability
- GENERAL_QUESTION : General questions about products/services
- CUSTOM_ORDER : Request for custom book orders

Validation Rules:

- Subject is required (max 200 characters)
- Message is required (max 2000 characters)
- Type must be valid enum value
- ProductId is optional (required for BOOK_AVAILABILITY)

2. List Inquiries

Endpoint: GET /api/inquiries

Description: Get paginated list of inquiries (filtered by user role)

Headers: Authorization: Bearer {token}

Query Parameters:

- page (int, default: 0): Page number
- size (int, default: 20): Page size

- `status` (String, optional): Filter by status (PENDING, IN_PROGRESS, RESOLVED, CLOSED)
- `type` (String, optional): Filter by type
- `productId` (Long, optional): Filter by product

Response: 200 OK

```
{
  "content": [
    {
      "id": 1,
      "type": "BOOK_AVAILABILITY",
      "subject": "Availability of Mathematics Class 10",
      "status": "PENDING",
      "productName": "Mathematics for Class 10",
      "userName": "John Doe",
      "createdAt": "2025-12-23T10:00:00Z",
      "hasResponse": false
    }
  ],
  "page": 0,
  "size": 20,
  "totalElements": 50,
  "totalPages": 3
}
```

Authorization:

- CUSTOMER: Only their own inquiries
- OWNER: All inquiries

3. Get Inquiry Details

Endpoint: GET `/api/inquiries/{id}`

Description: Get detailed information about a specific inquiry

Headers: Authorization: Bearer {token}

Path Parameters:

- `id` (Long): Inquiry ID

Response: 200 OK

```

{
  "id": 1,
  "type": "BOOK_AVAILABILITY",
  "subject": "Availability of Mathematics Class 10",
  "message": "Do you have Mathematics for Class 10 by Dr. Ramesh Kumar in stock?",
  "productId": 1,
  "productName": "Mathematics for Class 10",
  "status": "IN_PROGRESS",
  "userId": 5,
  "userName": "John Doe",
  "userEmail": "john@example.com",
  "createdAt": "2025-12-23T10:00:00Z",
  "updatedAt": "2025-12-23T11:00:00Z",
  "resolvedAt": null,
  "responses": [
    {
      "id": 1,
      "message": "Yes, we have this book in stock. You can visit our store or we can arrange deli",
      "userId": 1,
      "userName": "Store Owner",
      "createdAt": "2025-12-23T11:00:00Z"
    }
  ]
}

```

Authorization:

- CUSTOMER: Only their own inquiries
- OWNER: All inquiries

Error Responses:

- 404 Not Found : Inquiry not found
- 403 Forbidden : Not authorized to view this inquiry

4. Update Inquiry Status

Endpoint: PUT /api/inquiries/{id}/status

Description: Update inquiry status (owner only)

Headers: Authorization: Bearer {token}

Path Parameters:

- `id` (Long): Inquiry ID

Request Body:

```
{  
  "status": "RESOLVED"  
}
```

Response: 200 OK

```
{  
  "id": 1,  
  "status": "RESOLVED",  
  "resolvedAt": "2025-12-23T12:00:00Z",  
  "updatedAt": "2025-12-23T12:00:00Z"  
}
```

Authorization: OWNER only**Valid Status Transitions:**

- PENDING → IN_PROGRESS, CLOSED
- IN_PROGRESS → RESOLVED, CLOSED
- RESOLVED → CLOSED
- CLOSED → (final state)

5. Add Response to Inquiry

Endpoint: POST `/api/inquiries/{id}/response`

Description: Add owner response to an inquiry

Headers: Authorization: Bearer {token}

Path Parameters:

- `id` (Long): Inquiry ID

Request Body:

```
{
  "message": "Yes, we have this book in stock. You can visit our store or we can arrange deliver
}
```

Response: 201 Created

```
{
  "id": 2,
  "inquiryId": 1,
  "message": "Yes, we have this book in stock. You can visit our store or we can arrange deliver
  "userId": 1,
  "userName": "Store Owner",
  "createdAt": "2025-12-23T11:00:00Z"
}
```

Authorization: OWNER only

Business Rules:

- Automatically sets inquiry status to IN_PROGRESS if PENDING
- Sends email notification to customer
- Creates in-app notification

6. Get My Inquiries

Endpoint: GET /api/inquiries/my-inquiries

Description: Get current user's inquiries

Headers: Authorization: Bearer {token}

Query Parameters:

- page (int, default: 0): Page number
- size (int, default: 20): Page size
- status (String, optional): Filter by status

Response: 200 OK (Same structure as List Inquiries)

Authorization: Requires authentication

7. Get Pending Inquiries

Endpoint: GET /api/inquiries/pending

Description: Get all pending inquiries (owner only)

Headers: Authorization: Bearer {token}

Query Parameters:

- page (int, default: 0): Page number
- size (int, default: 20): Page size

Response: 200 OK (Same structure as List Inquiries)

Authorization: OWNER only

8. Get Inquiry Statistics

Endpoint: GET /api/inquiries/statistics

Description: Get inquiry statistics (owner only)

Headers: Authorization: Bearer {token}

Query Parameters:

- period (String, optional): Time period (TODAY, WEEK, MONTH, YEAR)

Response: 200 OK

```
{
  "total": 150,
  "pending": 25,
  "inProgress": 10,
  "resolved": 100,
  "closed": 15,
  "byType": {
    "BOOK_AVAILABILITY": 80,
    "GENERAL_QUESTION": 50,
    "CUSTOM_ORDER": 20
  },
  "averageResponseTime": "2.5 hours",
  "period": "MONTH"
}
```

Authorization: OWNER only

9. Update Inquiry

Endpoint: PUT /api/inquiries/{id}

Description: Update inquiry (customer can update their own, owner can update any)

Headers: Authorization: Bearer {token}

Path Parameters:

- id (Long): Inquiry ID

Request Body:

```
{
  "subject": "Updated subject",
  "message": "Updated message"
}
```

Response: 200 OK (Updated inquiry)

Authorization:

- CUSTOMER: Only their own inquiries (only if PENDING)
- OWNER: Any inquiry

Business Rules:

- Customers can only update PENDING inquiries
- Owner can update any inquiry

10. Delete Inquiry

Endpoint: DELETE /api/inquiries/{id}

Description: Delete an inquiry

Headers: Authorization: Bearer {token}

Path Parameters:

- id (Long): Inquiry ID

Response: 204 No Content

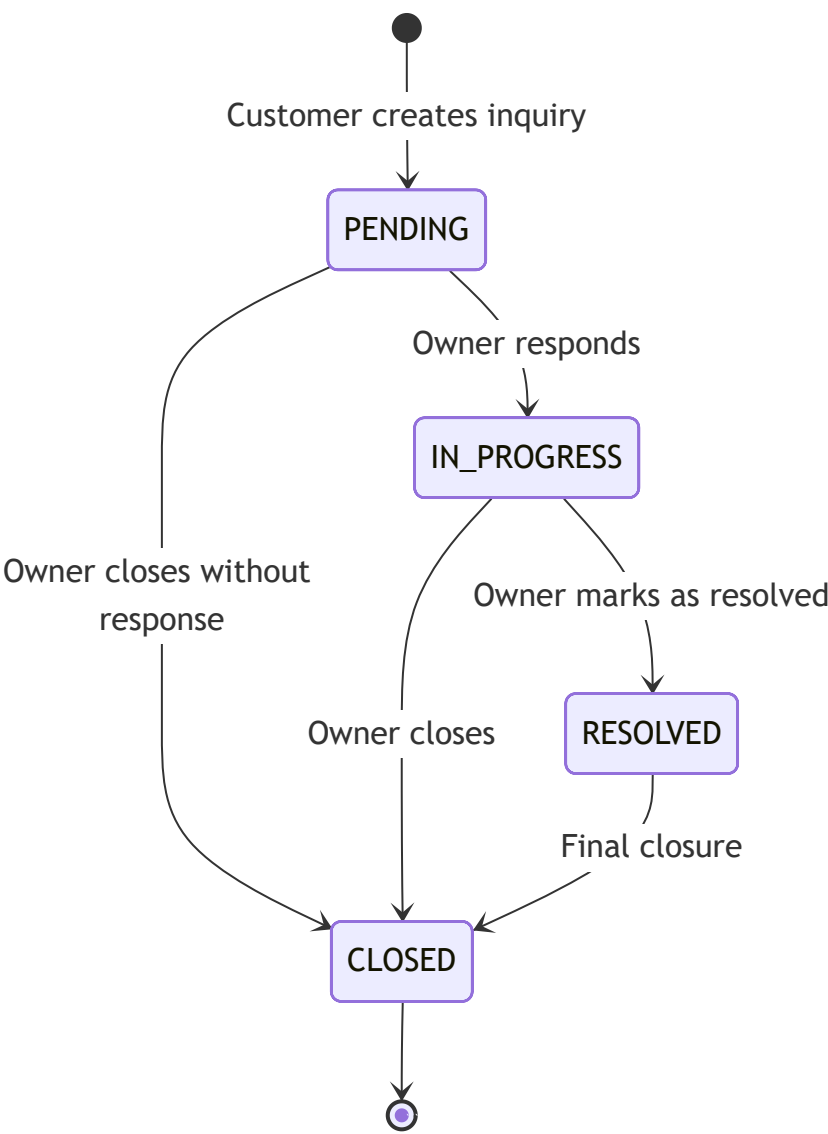
Authorization:

- CUSTOMER: Only their own inquiries (only if PENDING)
- OWNER: Any inquiry

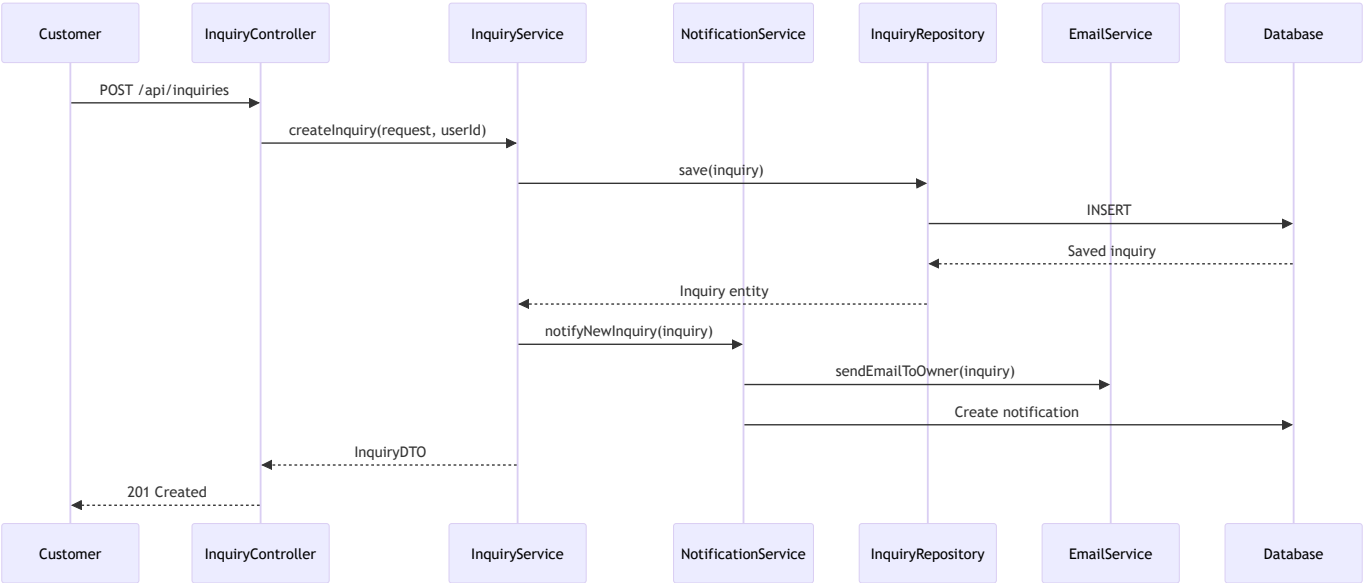
Business Rules:

- Customers can only delete PENDING inquiries
- Owner can delete any inquiry

Inquiry Lifecycle



Service Flow



Inquiry Entity

```
@Entity
@Table(name = "inquiries")
public class Inquiry {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @ManyToOne
    @JoinColumn(name = "user_id", nullable = false)
    private User user;

    @Enumerated(EnumType.STRING)
    @Column(nullable = false)
    private InquiryType type;

    @Column(nullable = false, length = 200)
    private String subject;

    @Column(nullable = false, columnDefinition = "TEXT")
    private String message;

    @ManyToOne
    @JoinColumn(name = "product_id")
    private Product product;

    @Enumerated(EnumType.STRING)
    @Column(nullable = false)
    private InquiryStatus status;

    @OneToMany(mappedBy = "inquiry", cascade = CascadeType.ALL)
    private List<InquiryResponse> responses;

    @CreationTimestamp
    private LocalDateTime createdAt;

    @UpdateTimestamp
    private LocalDateTime updatedAt;

    private LocalDateTime resolvedAt;
}
```

Inquiry Types

```
public enum InquiryType {  
    BOOK_AVAILABILITY,    // Inquiry about book availability  
    GENERAL_QUESTION,     // General questions  
    CUSTOM_ORDER          // Custom book orders  
}
```

Inquiry Status

```
public enum InquiryStatus {  
    PENDING,              // Newly created, awaiting response  
    IN_PROGRESS,          // Owner has responded, in progress  
    RESOLVED,              // Inquiry resolved  
    CLOSED                 // Inquiry closed  
}
```

Inquiry Response Entity

```
@Entity
@Table(name = "inquiry_responses")
public class InquiryResponse {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @ManyToOne
    @JoinColumn(name = "inquiry_id", nullable = false)
    private Inquiry inquiry;

    @ManyToOne
    @JoinColumn(name = "user_id", nullable = false)
    private User user;

    @Column(nullable = false, columnDefinition = "TEXT")
    private String message;

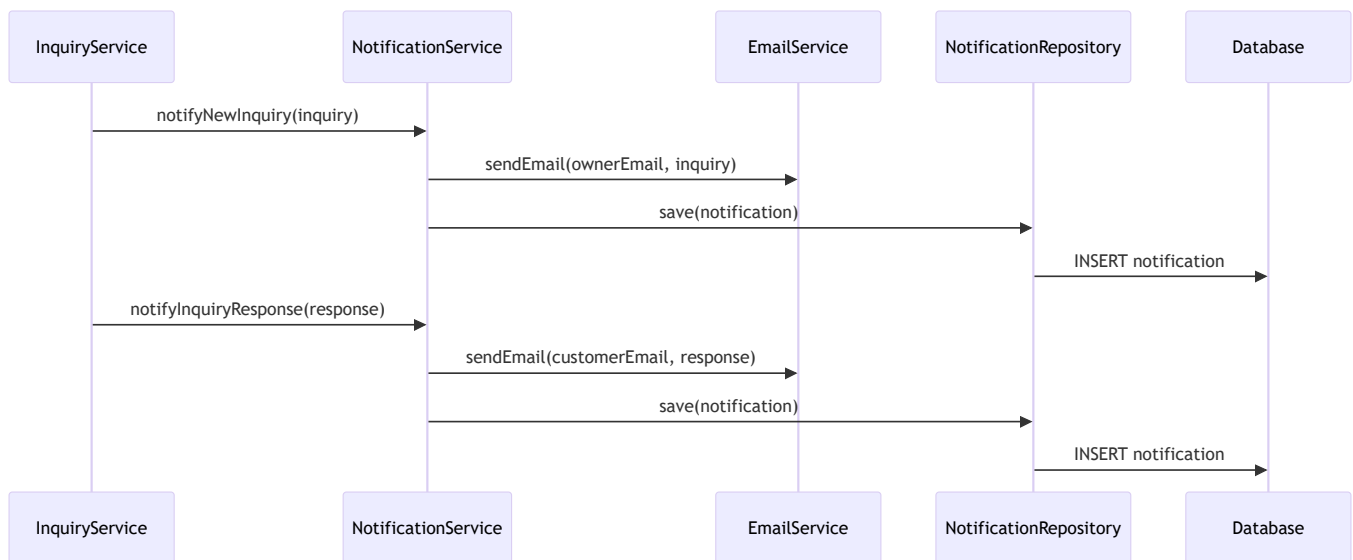
    @CreationTimestamp
    private LocalDateTime createdAt;
}
```

Service Methods

InquiryService Interface

```
public interface InquiryService {  
    InquiryDTO createInquiry(CreateInquiryRequest request, Long userId);  
    Page<InquiryDTO> getInquiries(InquiryFilter filter, Pageable pageable, User currentUser);  
    InquiryDTO getInquiryById(Long id, User currentUser);  
    InquiryDTO updateInquiryStatus(Long id, InquiryStatus status);  
    InquiryResponseDTO addResponse(Long inquiryId, String message, Long userId);  
    Page<InquiryDTO> getMyInquiries(Long userId, InquiryFilter filter, Pageable pageable);  
    Page<InquiryDTO> getPendingInquiries(Pageable pageable);  
    InquiryStatisticsDTO getStatistics(String period);  
    InquiryDTO updateInquiry(Long id, UpdateInquiryRequest request, User currentUser);  
    void deleteInquiry(Long id, User currentUser);  
}
```

Notification Flow



Database Queries

Optimized Queries

Get Inquiries with Filters:

```

SELECT
    i.id,
    i.type,
    i.subject,
    i.status,
    i.created_at,
    u.first_name || ' ' || u.last_name as user_name,
    p.name as product_name
FROM inquiries i
INNER JOIN users u ON i.user_id = u.id
LEFT JOIN products p ON i.product_id = p.id
WHERE
    (:status IS NULL OR i.status = :status)
    AND (:type IS NULL OR i.type = :type)
    AND (:productId IS NULL OR i.product_id = :productId)
    AND (:userId IS NULL OR i.user_id = :userId)
ORDER BY i.created_at DESC
LIMIT :size OFFSET :offset

```

Performance Considerations

Indexing

- **Primary Key:** id (auto-indexed)
- **Indexes:** userId, productId, status, type, createdAt
- **Composite Index:** (userId, status) for user inquiry queries
- **Composite Index:** (status, createdAt) for pending inquiries

Caching

- **Statistics Cache:** Cache inquiry statistics (5-minute TTL)
- **User Inquiry Cache:** Cache user's recent inquiries

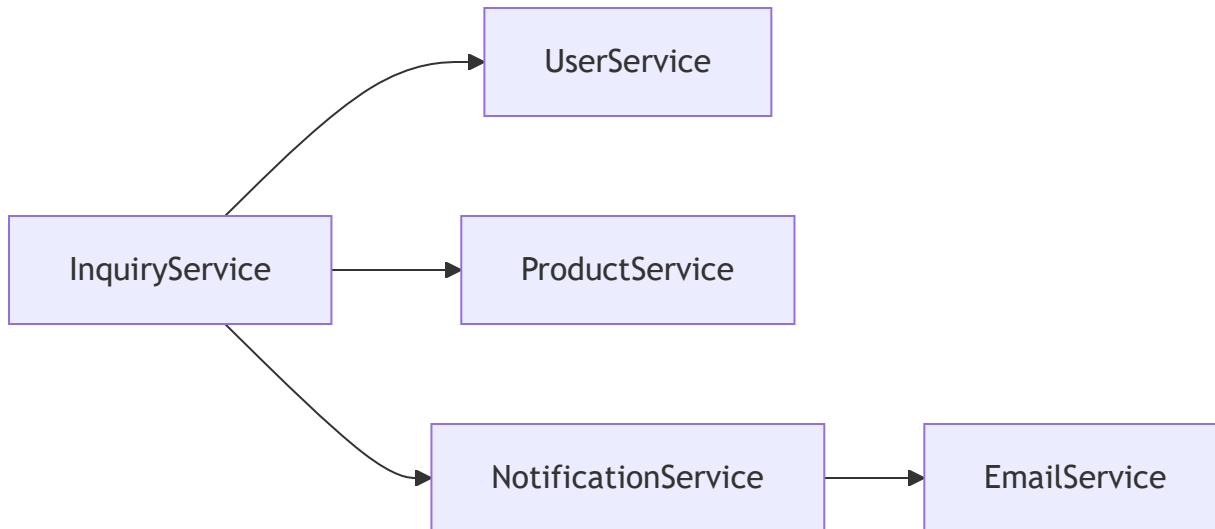
Error Handling

Custom Exceptions

- **InquiryNotFoundException:** When inquiry is not found
- **UnauthorizedInquiryAccessException:** When user tries to access unauthorized inquiry

- **InvalidInquiryStatusException**: When status transition is invalid
- **InquiryUpdateNotAllowedException**: When inquiry cannot be updated

Integration with Other Services



- Inquiry Service validates user and product existence
- Notification Service handles email and in-app notifications
- Product Service provides product details for inquiries

Testing Considerations

Unit Tests

- Inquiry creation with valid/invalid data
- Status transition validation
- Authorization checks
- Response creation

Integration Tests

- Complete inquiry lifecycle
- Notification triggering
- Email sending
- Statistics calculation

Business Rules

1. **Inquiry Creation:** Only authenticated customers can create inquiries
2. **Status Transitions:** Only valid status transitions allowed
3. **Response:** Only owner can respond to inquiries
4. **Update:** Customers can only update PENDING inquiries
5. **Delete:** Customers can only delete PENDING inquiries
6. **Notifications:** Email sent on new inquiry and response

Future Enhancements

- Inquiry priority levels
- Inquiry assignment to staff
- Inquiry templates
- Bulk inquiry operations
- Inquiry analytics dashboard
- Customer inquiry history
- Inquiry search functionality